

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

CMOS mixed signals VLSI design [ECE_4445]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With neat diagram explain the mixed signal design blocks (4)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Illustrate with example issues related with mixed signal design. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Explain the working of parasitic intensive switched capacitor integrator in sampling and integrate mode. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Design a logic to generate two non-overlapping clock signals has state sequence of 01-10-11-00. (4)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) With diagram explain the working of latched comparator. List the advantages. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Discuss the method to eliminate input offset voltage in two stage CMOS comparator. (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Find the number of input combinations, values for 1 LSB, and the full-scale voltage generated for a 3-bit, 8-bit, and 16-bit DAC, assuming that $V_{ref} = 5\text{ V}$. (4)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Determine the DNL for the 3-bit nonideal DAC whose transfer curve is shown in figure 3B. Assume that $V_{ref} = 5\text{ V}$ and Ideal height=1LSB. (3)

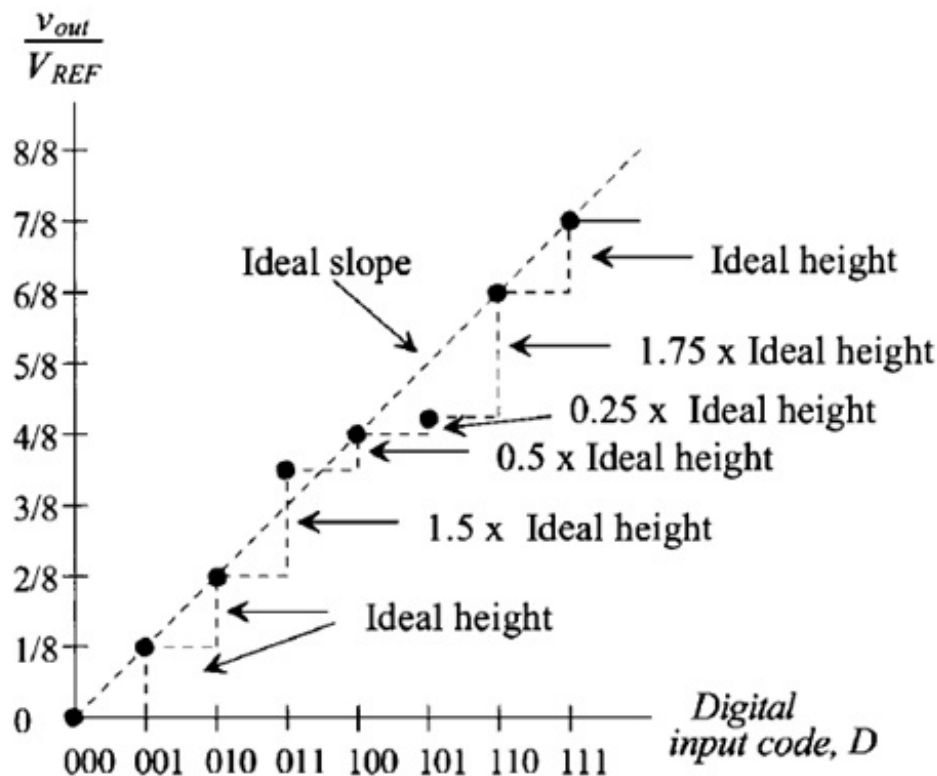


Figure 3B

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Design a 3-bit Flash converter, listing the values of the voltages at each resistor tap, and draw the transfer curve for $V_{in} = 0$ to 5 V. Assume $V_{ref} = 5$ V. Construct a table listing the values of the thermometer code and the output of the decoder for $V_{in} = 1.5$ V, 3.2 V, and 4.7 V. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Design a 3-bit charge-scaling DAC and find the value of the output voltage for $D_2D_1D_0 = 010$ and 101. Assume that $V_{ref} = 5$ V and $C = 0.5$ pF. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Explain the working of Type-II PLL with neat diagram. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Explain how charge Pump helps the PFD(phase frequency detector) to stabilize the voltage in PLL design (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Calculate the equivalent resistance of 5pF capacitance sampled at 100KHz clock frequency. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A 5μm/40nm MOSFET biased at 1mA exhibits a transconductance of 1/80 ohm . If the sheet resistance of the gate polysilicon is equal to 20ohm/square, what is the widest finger that the structure can incorporate while ensuring that the gate thermal noise voltage is 1/4th of the gate (3)

referred channel thermal noise voltage?

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Explain, how adding dummy fingers reduces the shallow trench isolation. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4445:CMOS mixed signals VLSI design

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	17
3	3	CO 3	14
4	4	CO 4	6
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4445:CMOS mixed signals VLSI design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)